

GameLink

A Gaming Clip Social Platform

CS335 - Software Engineering

Contribution Statement

- **Conor Gilligan 24395216:** Section 1 (Introduction), Section 2 (Stakeholders and Actors), and Section 3 (User Stories), Section 6 (UML Diagrams – Sequence Diagram) and Section 8 (System Testing)
- **Ray Gadliauskas 24463396:** Section 7 (User Interface Mockups), Section 6 (UML Diagrams – Use Case, Architecture) and Section 9 (Conclusion)
- **Sean Garland Prouse 24417742:** Section 4 (Functional Requirements), Section 5 (Non-Functional Requirements), Section 6 (UML Diagrams – Class Diagrams)

As well as preparation of the presentation slides and video

All members reviewed and improved the whole project before submitting.

1. Introduction

Gaming has grown a lot over the years and sharing short clips has become a big part of how people interact online. People spend hours of the day scrolling on their phone looking at videos of funny moments or random games they enjoy. Most of these clips are shared on Instagram or TikTok which aren't designed for gamers. There is no app built for just gaming clips.

It can be hard for some people to find gaming clips of specific games they like or connect with people who like the same games.

This project introduces GameLink which will be a platform made specifically for gamers. Users can upload gameplay clips, tag them with the game and watch clips from other users. They can also like, comment, follow and message people to make friends or teammates. There is also a moderator role to deal with reports or remove anything inappropriate and remove videos with the wrong tags.

The goal of this system is to make it easier for gamers to share videos or images and connect with each other all in the same space.

2. Stakeholders and Actors

- **Gamers:** The main users. They upload clips, watch content and interact with each other.
- **Moderators:** Moderators are responsible for reviewing reported clips and making sure the platform stays safe
- **Platform Admins:** These users manage the system overall, including maintenance and policies.

3. User Stories

1. As a gamer I want to create an account where I can upload my clips.
2. As a gamer I want to log in securely so that I can access my profile and saved settings.
3. As a gamer I want to browse clips by game so that I can watch content from games I enjoy.
4. As a gamer I want to add my favourite games to my profile so that other users can see what I play
5. As a gamer I want to save clips I like so i can rewatch them
6. As a moderator I want to view a history of reports for a user so that repeated harmful behaviour can be identified.
7. As a moderator I want to remove offensive comments and clips
8. As a moderator I want to suspend users who repeatedly break rules so that the community stays respectful
9. As a Moderator I want to be notified when reports are made so I can view them
10. As a moderator I want to see the reason a clip or user was reported so that I can make the right decision.

The user stories were made by importance, basic features like creating an account, logging in and uploading clips were the most important. Other features like saving clips and moderation tools were included after to improve the system.

The user stories were also used to help with system tests. It helped make sure that everything the user needs is included.

4. Functional Requirements

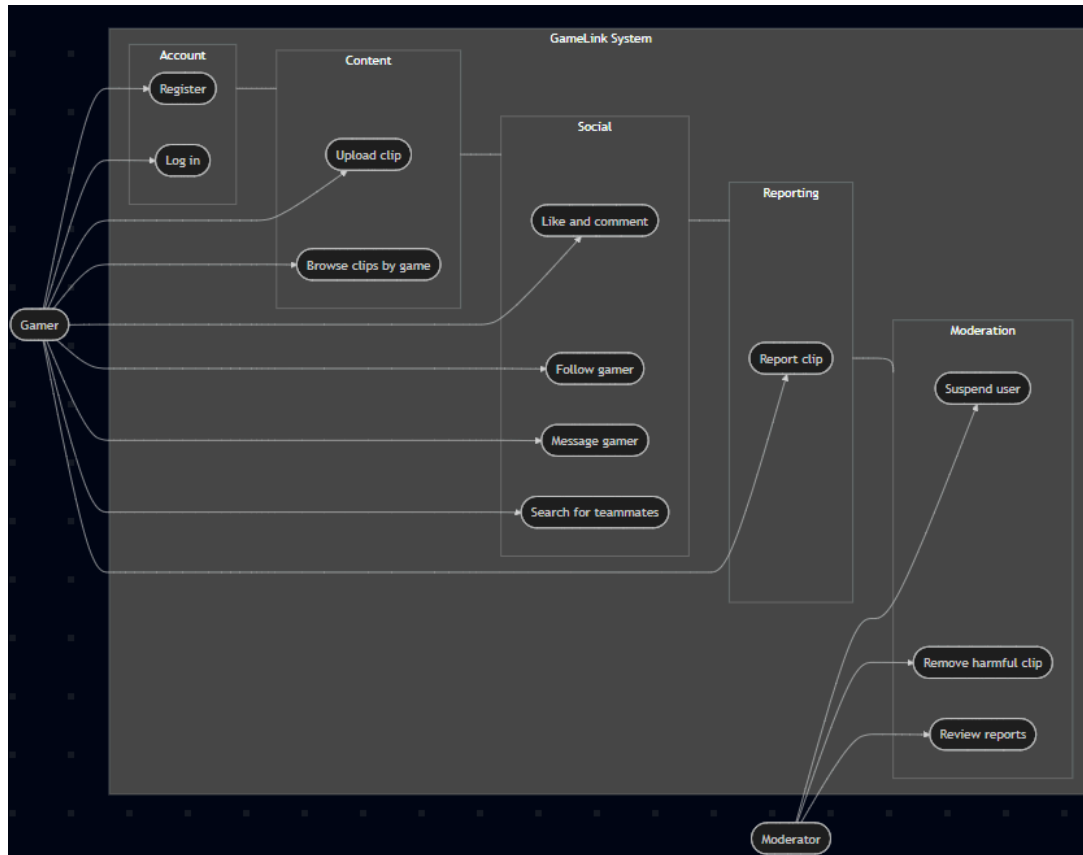
- Users Create accounts
- Users can edit their profile
- Users can upload and watch clips
- Clips can be tagged with games
- Users can search for clips or tags
- Users can like comment and follow
- Users can report clips or profiles
- Moderators can review reports
- Moderators can remove clips and Users

5. Non-Functional Requirements

- Clips should upload quick
- The system should handle many users at the same time.
- Videos should look good on phone and computer (no lag)
- User data stored securely
- App easy to use
- App handle a lot of people
- Reports reviewed ASAP

6. UML Diagrams

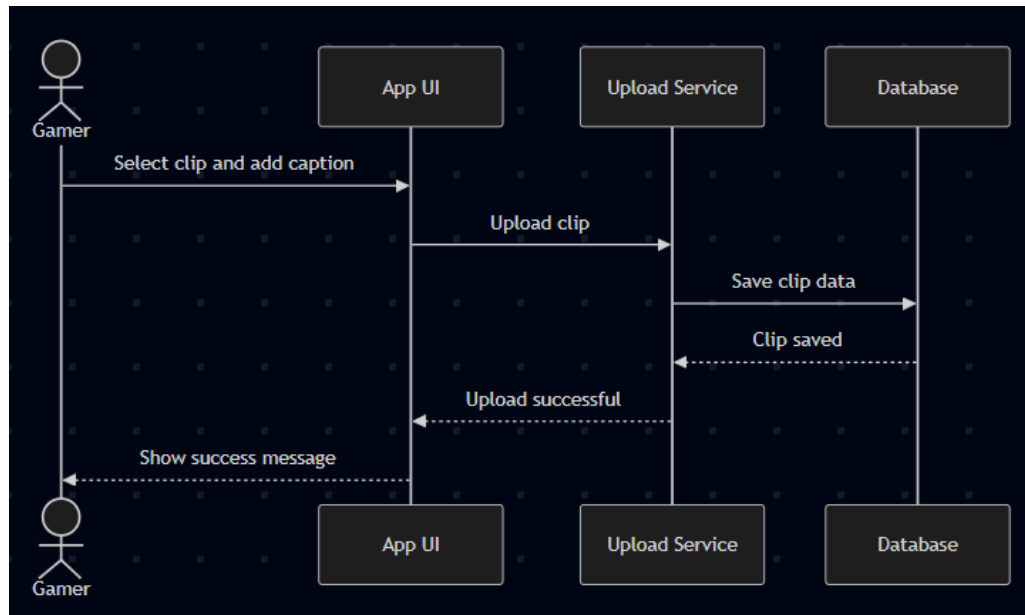
USE CASE DIAGRAM



This diagram was made on mermaid live. The use case diagram shows the 2 main actors in the system, Gamer and Moderator, The gamer performs social and content sharing actions while moderators handle reporting and enforcement.

The diagram shows how users interact with the system. It shows the difference between gamers and moderators and what they can do. This helps make the system easier to manage.

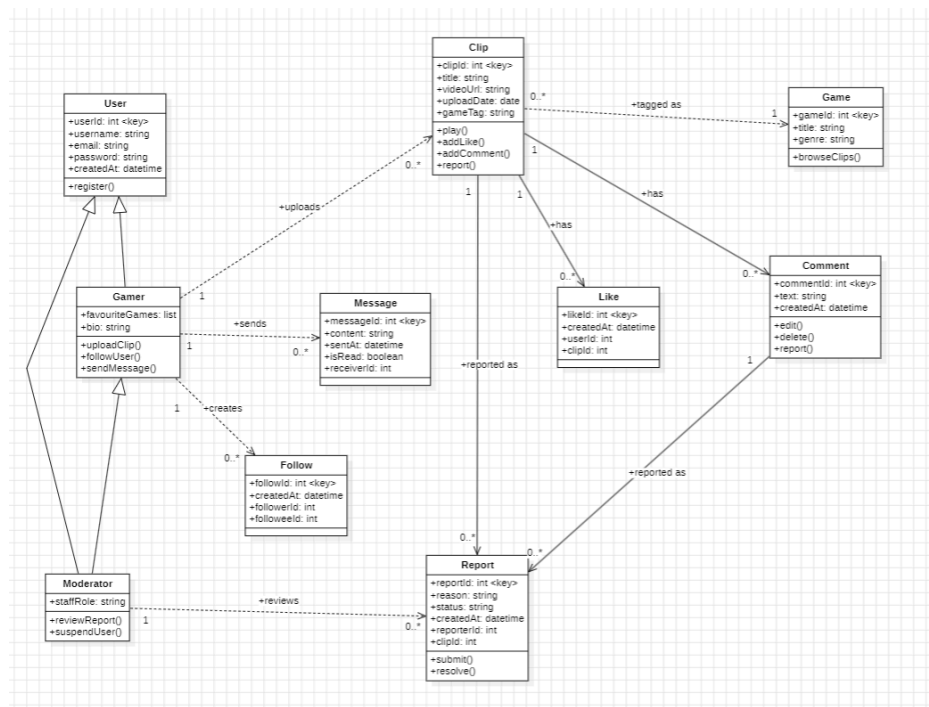
SEQUENCE DIAGRAM



This diagram was created on Mermaid Live, it models the upload process. A gamer selects a clip and the upload service validates it, it is stored in the database and the upload service publishes the clip so it becomes visible to other users.

It shows the steps to uploading, it shows the frontend, backend and database working together. This makes sure the system works well and avoids errors.

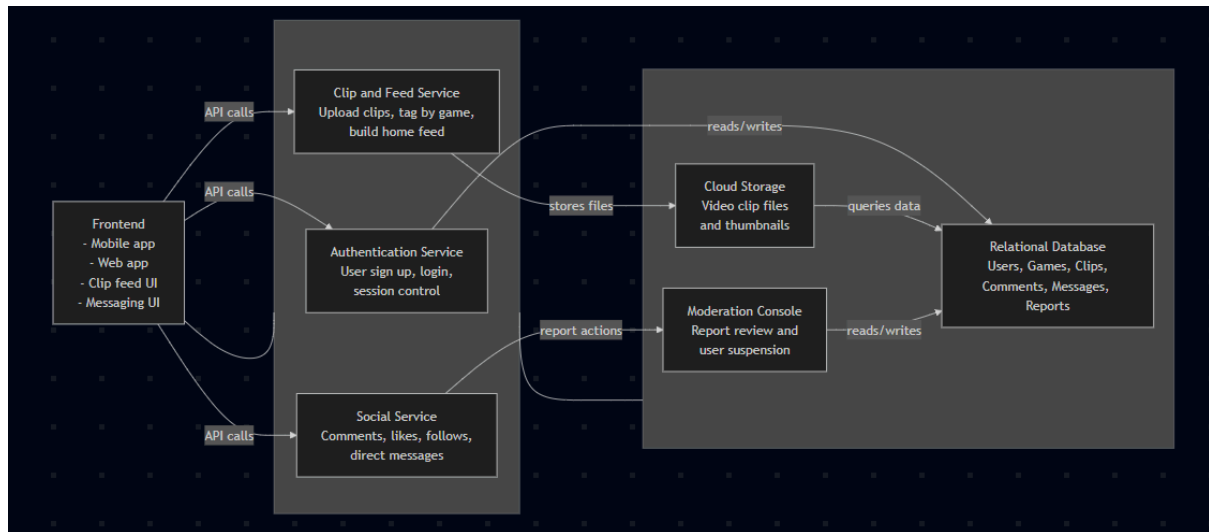
CLASS DIAGRAM



The class diagram made using StarUML, identifies the main objects in the system and how they relate to each other. User is the parent class with gamer and moderator as specialised roles. Clips, games, comments, follows, etc, represent the platform's main content.

It shows how gamers and moderators are different but are still related.

ARCHITECTURE DIAGRAM



This was also made using mermaid live. This diagram uses a three layer view. The front end provides the app screen. The backend manages authentication and social actions. The storage keeps data and video files.

The frontend is what the user sees, the backend handles logic and the database stores all its data.

7. UI Mockups



Valorant

♡ 12.5K 💬 342

@valorantace

Insane 1v5 clutch



Home/Feed screen:

This screen shows the main feed where users can scroll through clips posted by others. Each clip displays the game, username, caption, and engagement like likes and comments. It works similar to social media apps where you just scroll and discover content.

 Search games or clips...

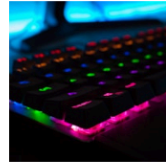
Popular Games



Valorant
2.3K clips



FIFA 24
1.8K clips



Minecraft
3.1K clips



Call of
2.7K clip

Trending Clips



Explore/Search screen:

This screen lets users search for games or clips and discover popular content. It shows trending games and clips so users can easily find what's popular. There is also a search bar at the top to quickly look up specific games or videos.

Select video

Title

Select game

Post

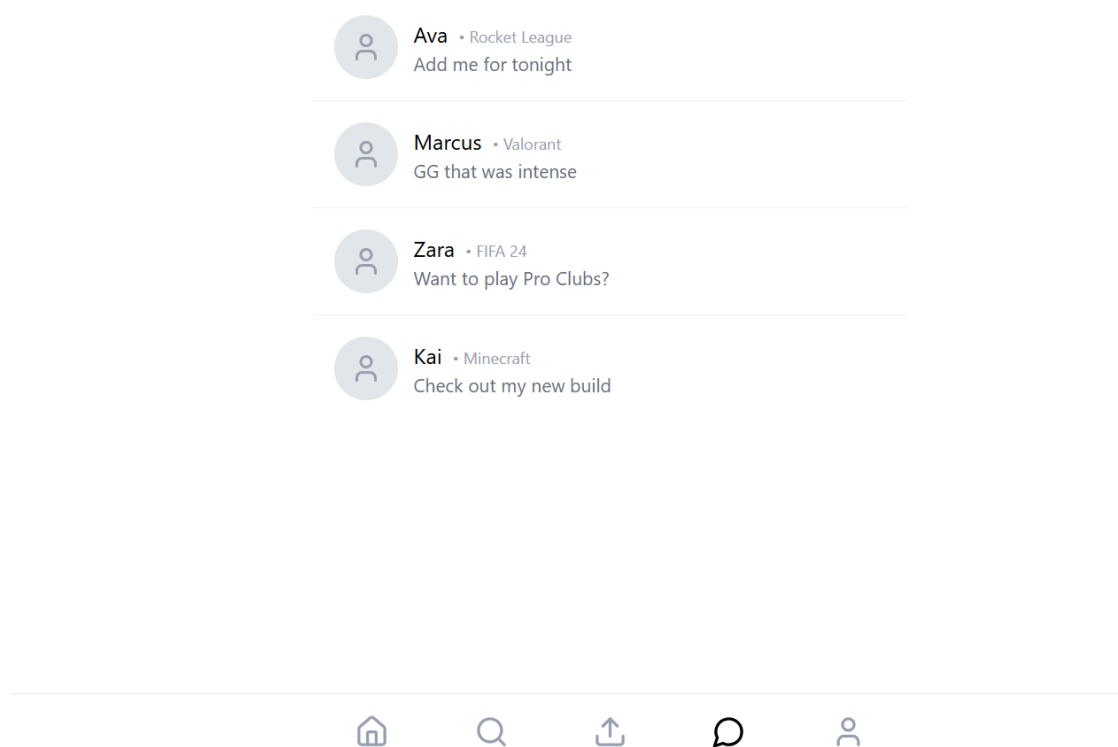


Upload Clip screen:

This is where users upload their own gameplay clips.

They can select a video, add a title, choose the game, and then post it.

It's a simple form layout to make uploading quick and easy.

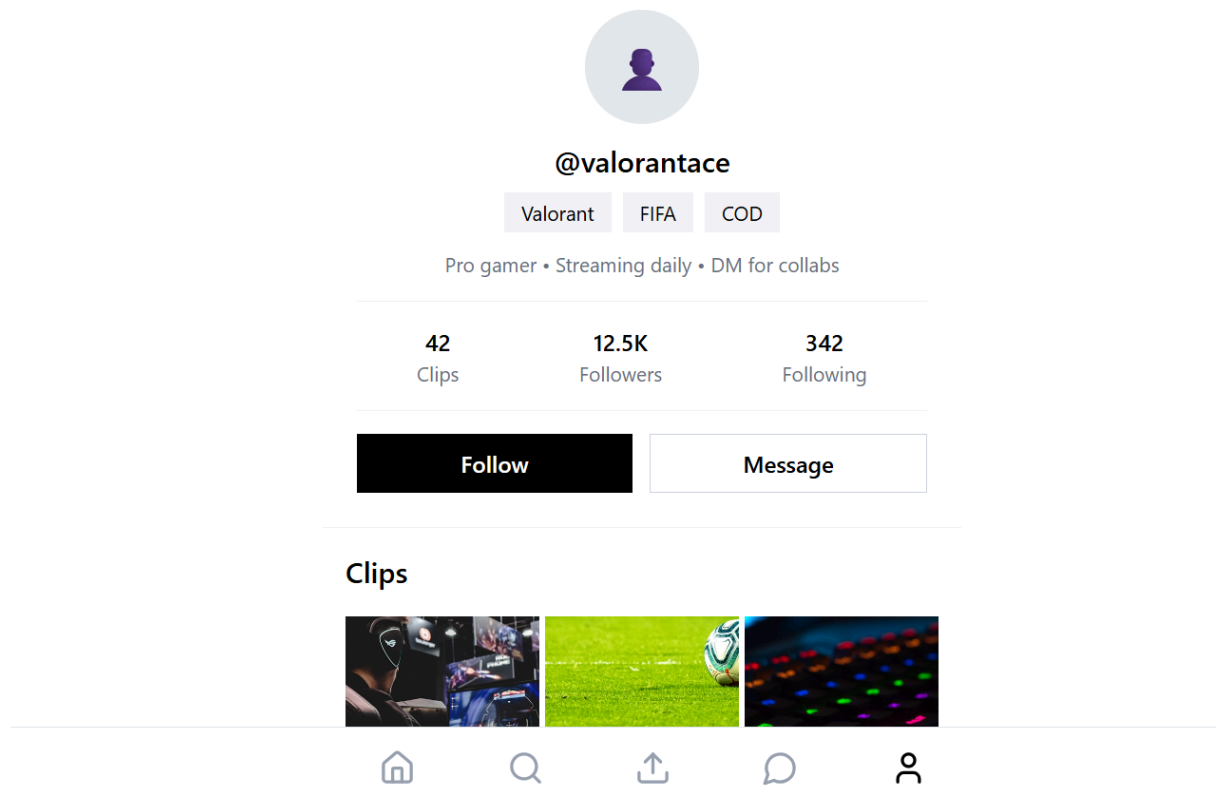


Messaging screen:

This screen shows conversations with other users.

Users can see recent messages and open chats to talk to teammates or friends.

It helps players communicate and organise games together.



User profile screen:

This screen displays a user's profile, including their clips and stats.

You can see how many clips they've posted, their followers, and what games they play.

There are also buttons to follow the user or send them a message.

8. System Testing

The following test cases check whether the system satisfies the main requirements.

Table of system testing

TEST CASE	DESCRIPTION	INPUT	EXPECTED RESULT
TC1	Register	Username, Email, Password	Account is created
TC2	Log in	Email and Password	User reaches home page
TC3	Upload Clip	Video, Title, Caption	Clip is stored and appears on feed
TC4	Browse Games	User selects game	Only clips of that game are shown
TC5	Send Message	User sends direct message	User receives message
TC6	Like and Comment	User taps like and comment	Like count updates and comment is displayed
TC7	Report Clip	User submit report	Report is stored for moderator to review
TC8	Moderator Removes clip	Moderator removes the clip	Clip no longer appears in feeds
TC9	Invalid Login	Wrong email	Error message
TC10	Upload Error	Wrong file type	Upload rejected
TC11	Missing Information	Empty info during signup	Asked to complete signup

9. Conclusion

In conclusion GameLink is an app that was designed to provide a platform specifically for gamers to share, discover and involve themselves with gameplay content. Unlike the many other existing social media apps such as Tiktok or Instagram, GameLink is only built around the needs of the gaming community and offers features like viewing gameplay clips, communication features in the form of text messages and video calls, and personalised user profiles, all while being fully moderated to protect the safety of users.

Throughout the development of this project, we identified the necessary parts needed to design and document the different systems in the app through stakeholder/actor analysis, user stories, functional and non functional requirements, and UML diagrams. These diagrams, including the use case, sequence, class and architecture diagrams, provide a clear picture of how the system is structured and how users will interact with it. The UI mockups further show how the app will look and feel on mobile.

The system testing table shows that the platform's key features such as, registration/login, clip uploading, moderation and messaging can be verified through clear test cases. This shows that the app will meet user expectations and be reliable under normal conditions.

Overall, GameLink has potential to become a one of a kind platform within the gaming community. With continued updates in the future and a decent moderation team, it could become an extremely popular space for gamers to share their best clips and interests.